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Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

What is this?

This is the course material and a template designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL** to create interactive teaching books easily. The main idea of this template is for it to be broad enough to cover many use cases and for each student to adapt it to their own course, while also being sufficiently equipped to be used easily with the help of AI assistants (such as GitHub Copilot, Gemini, Claude, Codex, etc.) and a code editor such as VS Code.

Content

In this book you will find:

- Tutorials to learn how to use the template
- Examples by Degree to see real cases
- Information on how to cite and licenses

PDF version

You can also download the printable version of the book:

- [Download PDF in English](#)
- [Descargar PDF en español](#)

Note

This project is designed to be used with **VS Code** and **AI** assistants.

Part I

Presentation

AUTHOR PRESENTATION

This book has been prepared by Fernando Espejo Almodóvar.

1.1 Short Bio

Fernando Espejo Almodóvar is a Tenured University Professor at the University of Salamanca (USAL), affiliated with the Higher Polytechnic School of Ávila. He is an expert in civil engineering and water resources management.

Among his academic responsibilities, he serves as deputy director of the Master's Degree in Water Systems Modeling at USAL. His research focuses on the integral water cycle, risk management in irrigation infrastructures, hydrology and computational hydraulics, as well as the application of artificial intelligence models in dam engineering.

Part II

Introduction

IMPORTANCE OF REGULATING DAMS

Regulating dams are fundamental infrastructures in water resources management at the basin level. Their primary function is to store water during periods of abundance (heavy rainfall or snowmelt) so that it can be distributed in a controlled manner during times of scarcity or drought.

2.1 Benefits in Basin Management

The proper planning and operation of these dams allow achieving multiple key objectives for sustainable development and safety:

1. **Supply Guarantee:** They ensure a continuous supply of drinking water for urban areas, as well as for agricultural irrigation and industrial use.
2. **Flood Control:** They act as buffering elements during torrential rains, significantly reducing the risk of flooding downstream.
3. **Energy Generation:** Many of these infrastructures integrate hydroelectric power plants, providing a renewable and manageable source of energy.
4. **Maintenance of Ecological Flows:** They allow regulating the released flows to ensure the survival of river ecosystems even during dry seasons.

2.2 The Modeling Challenge

Managing a river basin with multiple dams requires complex analysis. Water release decisions must be based on precise hydrological and hydraulic models, decision support tools, and increasingly, artificial intelligence algorithms. Throughout this book, we will explore the theoretical foundations and tools to understand and optimize these vital systems.

2.3 Common Dam Typologies

To understand regulation, it is useful to distinguish between the predominant structural designs. In [Fig. 2.1](#) we observe the typical appearance of a concrete gravity dam, very common in narrow valleys. On the other hand, [Fig. 2.2](#) illustrates an embankment dam, commonly used when the foundation is more deformable or the valley is wider.



Figure2.1: Concrete gravity dam in a mountainous environment.



Figure2.2: Embankment dam holding back a large reservoir.